Group 2: Pima Diabetes Study

Dataset Guidance

Dataset Overview

File: `pima_diabetes.csv`

Sample Size: n = 768 Pima Indian women

Research Context: Cross-sectional study examining risk factors for diabetes. Multiple health measures collected to identify which factors are associated with diabetes diagnosis.

Variables

- `glu`: Plasma glucose concentration (mg/dL)
- `bmi`: Body mass index (kg/m^2)
- `age`: Age (years)
- `type`: Diabetes diagnosis (No or Yes)
- Additional variables may include: blood pressure, pregnancies, skin fold thickness

Loading the Data in R

```r
# Load the dataset
pima <- read.csv("pima_diabetes.csv")

# View structure
str(pima)
head(pima)

# Summary by diabetes status
table(pima$type)
tapply(pima$glu, pima$type, summary)
tapply(pima$bmi, pima$type, summary)
```

Research Question to Consider

"Which health factors are associated with diabetes?"

Alternatively: "Do glucose levels, BMI, and age differ between women with and without diabetes?"

Key Decision Point

The Challenge: You have a **binary outcome** (diabetes: Yes/No) and several continuous predictors (glucose, BMI, age). How do you analyze this with methods covered in this course?
Questions to explore with AI:

- What methods are commonly used for binary outcomes?
- Can you compare groups (Yes vs. No) on each predictor separately?
- What are the trade-offs between simple and complex approaches?

Important note: AI tools will likely suggest advanced methods for binary outcomes that you haven't learned yet. Consider whether simpler approaches can still answer your research question.

Things to Think About

Exploratory Analysis:

- Create visualizations comparing diabetes groups (boxplots, histograms)
- Calculate summary statistics by diabetes status
- Consider: Which variables show the biggest differences between groups?

Statistical Methods:

- Review methods for comparing groups
- Consider: How do you compare two groups on a continuous variable?
- Consider: What if you wanted to categorize continuous variables?

Multiple Predictors:

- You have several potential risk factors (glucose, BMI, age, etc.)
- Should you analyze them one at a time or together?
- What methods from this course allow you to examine each factor?

AI Consultation Strategy

Very Important: When you ask AI about analyzing binary outcomes (Yes/No), it will almost certainly suggest methods beyond introductory biostatistics.

Prompts to try:

- "How should I analyze which factors are associated with diabetes?"
- "I have a binary outcome and continuous predictors - what methods can I use?"
- "I haven't learned [method AI suggested]. Can I answer my question with methods covered in this course?"
- "What's the difference between analyzing predictors one at a time vs. together?"

Watch for:

- Tools suggesting methods you haven't encountered (especially for binary outcomes)
- Different recommendations from different AI tools
- Trade-offs between simple (univariate) and complex (multivariate) approaches

Critical thinking: Just because AI suggests an advanced method doesn't mean simpler methods can't answer your question. Can you identify risk factors using methods you've learned?

Questions for Your Analysis

As you work through this dataset, consider:

1. What research question are you answering?
2. Which statistical method(s) from this course can address this question?
3. Are you analyzing predictors individually or together? Why?
4. What are the limitations of your chosen approach?
5. How will you justify your methodological choice when AI suggested something different?

Interpretation Considerations

If you find significant differences between diabetes groups:

- What does this tell you about risk factors?
- How would you interpret findings for clinical practice?
- What can you conclude from examining factors one at a time?
- What questions remain that would require more advanced methods?

Resources

- Week 9 drop-in session (Wed 12-1:20pm, 153 Milam)
- Textbook: Chapter 11 (Two-Sample Tests), Chapter 15 (Chi-Square)
- Your team!
- Canvas discussion board

Remember: Simple methods that you understand are often better than complex methods you don't. The goal is to:

- Answer your research question with appropriate methods from this course
- Understand and justify your analytical choices
- Critically evaluate what AI tools suggested and why
- Recognize when advanced methods exist but aren't necessary